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Wrench with work piece locking mechanism

Abstract

A wrench may include a head, a drive tang that engages a work piece, a locking detent operably coupled to the drive tang, and a release button configured to control operation of the locking detent. The release button may be movable between a locked position where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang. The wrench may also include a slide lock member configured to slide along a second direction between an engaged position and a disengaged position. In the engaged position, the slide lock member may prevent movement of the release button, and, in the disengaged position, the slide lock member may be disengaged from the release button to permit movement of the release button from the locked position to the unlocked position.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5386747	12/1994	Grover	81/177.85	B25B 23/0035
6044730	12/1999	Roberts et al.	N/A	N/A
6205890	12/2000	Roberts	81/177.85	B25B 23/0035
6752046	12/2003	Lee	81/177.85	B25B 13/461
8943929	12/2014	Ou	81/60	B25B 13/463
9205538	12/2014	Yang	N/A	B25B 13/461
9849569	12/2016	Su	N/A	B25B 23/0071
2010/0132517	12/2009	Crawford	N/A	N/A
2015/0246434	12/2014	Shyu	N/A	N/A
2016/0229042	12/2015	Shyu	N/A	N/A
2017/0190029	12/2016	Cuesta	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2264444	12/1998	CA	N/A
205111698	12/2015	CN	N/A
205111700	12/2015	CN	N/A
210210167	12/2019	CN	N/A
111070145	12/2019	CN	N/A
2002034474	12/2001	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion of corresponding International Application No. PCT/US2021/028146, mailed on Jul. 21, 2021, all enclosed pages cited herein. cited by applicant
Chinese Office Action and Search Report (with English Translation) issued in corresponding Chinese Patent Application No. 202180031121.0 mailed Aug. 20, 2024, all pages cited in its entirety. cited by applicant

Second Office Action (with English Translation) issued in corresponding Chinese Patent Application No. 202180031121.0 on Mar. 3, 2025, all pages cited in its entirety. cited by applicant

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Background/Summary

TECHNICAL FIELD
(1) Example embodiments generally relate to hand tool technology, and in particular to coupling technologies for wrenches that engage with work pieces, such as, for example, sockets.
BACKGROUND
(2) Wrenches, including ratchet wrenches, that have the ability to attach a drive tang to a variety of work pieces, such as sockets or bits, provide substantial application flexibly. For example, the same ratchet wrench may be attached to many different sockets to permit the single wrench to be utilized when installing or removing a variety of differently sized fasteners. However, the removable nature of such work pieces from the wrench creates a risk that the work piece could unintentionally be separated from the wrench. In this regard, for example, if a user is working at a height (e.g., on a ladder) or over a machine (e.g., over a car engine), the accidental and unintentional separation of the work piece from the wrench could lead to the work piece falling from the height and being lost, or falling into the machine and being difficult to retrieve. As such, there continues to be a need for new, secure and efficient mechanisms to prevent the accidental, unintentional separation of work pieces, such as sockets, from wrenches.
BRIEF SUMMARY OF SOME EXAMPLES
(3) As such, according to some example embodiments, a ratchet wrench is provided. The ratchet wrench may comprise a head comprising a top face and a bottom face, and a drive tang that extends from the bottom face of the head. The drive tang may be shaped to engage a work piece. The ratchet wrench may further comprise a ratcheting mechanism disposed within a head cavity of the head and operably coupled to the drive tang. Additionally, the ratcheting mechanism may be configured to permit the drive tang to rotate relative to the head in a ratcheting rotational direction and prevent the drive tang from rotating relative to the head in a drive rotational direction. The ratchet wrench may further comprise a locking detent operably coupled to the drive tang, and a release button disposed on a top face of the head and configured to control operation of the locking detent. The release button may be movable along a first direction between a locked position, where the locking detent locks the work piece to the drive tang, and an unlocked position where the detent permits removal of the work piece from the drive tang. The ratchet wrench may further comprise a slide lock member configured to slide along a second direction between an engaged position and a disengaged position. In this regard, the second direction may be perpendicular to the first direction. Additionally, in the engaged position, the slide lock member may engage with the release button to prevent movement of the release button in the first direction from the locked position to the unlocked position, and, in the disengaged position, the slide lock member may be disengaged from the release button to permit movement of the release button in the first direction to move from the locked position to the unlocked position.
(4) According to some example embodiments, a wrench is provided. The wrench may comprise a head comprising a top face and a bottom face, and a drive tang that extends from the bottom face of the head. In this regard, the drive tang may be shaped to engage a work piece. The wrench may further comprise a locking detent operably coupled to the drive tang and a release button disposed

on a top face of the head and configured to control operation of the locking detent. The release button may be movable along a first linear direction between a locked position where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang. The wrench may further comprise a slide lock member configured to slide along a second linear direction between an engaged position and a disengaged position. In this regard, in the engaged position, the slide lock member may engage with the release button to prevent movement of the release button in the first linear direction from the locked position to the unlocked position, and, in the disengaged position, the slide lock member may be disengaged from the release button to permit movement of the release button in the first linear direction to move from the locked position to the unlocked position.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWING(S)

(1) Having thus described some example embodiments in general terms, reference will now be made to the accompanying drawings, which are not necessarily drawn to scale, and wherein:

(2) FIG. **1** illustrates a side view of a wrench according to some example embodiments;

(3) FIG. **2** illustrates an exploded view of a wrench according to some example embodiments;

(4) FIG. **3**A illustrates a top perspective view of a slide lock member according to some example embodiments;

(5) FIG. **3**B illustrates a bottom perspective view of a slide lock member according to some example embodiments;

(6) FIG. **3**C illustrates a securing pin according to some example embodiments;

(7) FIG. **4**A illustrates a top view of a wrench with a slide lock member in an engaged position according to some example embodiments;

(8) FIG. **4**B illustrates a side view of a wrench with a slide lock member in an engaged position according to some example embodiments;

(9) FIG. **4**C illustrates a zoomed top view of a head of a wrench with a slide lock member in an engaged position according to some example embodiments;

(10) FIG. **4**D illustrates a cross-section view taken at A-A of FIG. **4**C of select components of a wrench including a slide lock member in an engaged position according to some example embodiments;

(11) FIG. **5**A illustrates a top view of a wrench with a slide lock member in a disengaged position according to some example embodiments;

(12) FIG. **5**B illustrates a side view of a wrench with a slide lock member in a disengaged position according to some example embodiments;

(13) FIG. **5**C illustrates a side view of a wrench with a slide lock member in a disengaged position and a release button in an unlocked position according to some example embodiments;

(14) FIG. **5**D illustrates a zoomed top view of a head of a wrench with a slide lock member in a disengaged position and the release button in an unlocked position according to some example embodiments;

(15) FIG. **5**E illustrates a cross-section view taken at B-B of FIG. **5**D of select components of a wrench including a slide lock member in a disengaged position according to some example embodiments; and

(16) FIGS. **6**A to **6**C provide an illustration of a two-step disengagement and unlocking process with respect to removal of a socket according to some example embodiments.

DETAILED DESCRIPTION

(17) Some example embodiments now will be described more fully hereinafter with reference to the accompanying drawings, in which some, but not all example embodiments are shown. Indeed,

the examples described and pictured herein should not be construed as being limiting as to the scope, applicability or configuration of the present disclosure. Rather, these example embodiments are provided so that this disclosure will satisfy applicable legal requirements. Like reference numerals refer to like elements throughout. Furthermore, as used herein, the term “or” is to be interpreted as a logical operator that results in true whenever one or more of its operands are true. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other. As used herein, operable coupling should be understood to relate to direct or indirect connection that, in either case, enables functional interconnection of components that are operably coupled to each other.

(18) According to various example embodiments, a wrench, such as a ratchet wrench, is provided with a work piece locking mechanism that is configured to avoid unintentional separation of a work piece from a drive tang of the wrench. As mentioned above, a work piece may be a socket, a bit, or the like that can be attached to a drive tang of a wrench to facilitate engagement of the wrench with a variety of fasteners. In this regard, the work piece locking mechanism may being configured to lock the work piece to the drive tang such that the work piece cannot be unintentionally removed from the drive tang by bumping or otherwise applying a force directly onto the work piece. In this regard, according to some example embodiments, the wrench may include a lockable detent that may engage and lock a work piece onto the drive tang. According to some example embodiments, a work piece may include a locking feature, such as an internal groove, that may engage with a locking member of the lockable detent that extends away from the drive tang to lock the work piece onto the drive tang.

(19) To release the work piece from the drive tang of the wrench, a two-step or two-movement action may be required to be performed by the user on the wrench. In this regard, according to some example embodiments, a wrench may include a release button that can be depressed to control the lockable detent. The release button may operate to release the locking member, which may be a component of the locking detent, from the work piece to permit separation of the work piece from the wrench. In this regard, the release button may be actuated between a locked position, where the locking member is in locked engagement with the work piece, and an unlocked position, where the locking member may be disengaged from the work piece and separated from the wrench. Actuation of the release button may occur along a locking direction in a linear path.

(20) Additionally, the wrench may comprise a slide lock member that selectively engages with the release button to affect the release button's ability to move. In this regard, the slide lock member may be movable, along an engagement direction, between an engaged position, where the slide lock member is engaged with the release button, and a disengaged position, where the slide lock member is not engaged with the release button. Accordingly, when the slide lock member is engaged with the release button, the release button may be prevented from moving into the unlocked position, and, when the slide lock member is in the disengaged position, the release button may be permitted to move into the unlocked position.

(21) To separate a work piece from a drive tang of a wrench, according to some example embodiments, a user may be required to first move the slide lock member from the engaged position to the disengaged position along a first (engagement) direction, as a first step or first movement, which may be a linear movement. Subsequently, the user may be required, while the slide lock member is maintained in the disengaged position, to press the release button to move the release button along a second (locking) direction from the locked position to the unlocked position, which also may be a linear movement and may be perpendicular to the movement of the slide lock member. Upon moving the release button into the unlocked position (no longer being obstructed or encumbered by the slide lock member due to the slide lock member being in the disengaged position), the locking member may be permitted to disengage from the work piece, to permit the work piece to the separated from the drive tang.

(22) The two-step or two-movement operation performed by the user to separate the work piece from the wrench may require, according to some example embodiments, movements by the user that are liner in different directions, such as in perpendicular directions. Such sequenced, movements are very unlikely to occur unintentionally. As such, by requiring a two-step or two-movement operation by the user to unlock a work piece, it is very unlikely that a work piece may be accidentally separated from a wrench without a user intending for the work piece to be removed from wrench. Accordingly, the collaborative operation of the slide lock member with the release button may provide a secure and efficient approach for controlling the removal or separation of a work piece, such as a socket or a bit, from the drive tang of a wrench, according to some example embodiments.
(23) Having described some aspects of some example embodiments in a general sense, FIG. **1** provides an example embodiment in the form of an example ratchet wrench **10**. Although the example embodiments may be described with respect to a ratcheting wrench **10**, it is understood that some example embodiments may be applicable to, for example, other non-ratcheting wrenches. In this regard, the wrench **10** may be configured to ratchet in a first rotational direction about an axis **205** or apply a torque to, for example, a fastener (e.g., a bolt, a nut, or the like) in a second, opposite, rotational direction, based on a position of an reversing lever **25** of the wrench and a direction of rotation applied by a user. As such, the wrench **10** may be configured to permit a user to apply torque on a fastener in either a clockwise or counterclockwise manner, with an ability to ratchet in an opposite direction of rotation based on the position of the reversing lever **25**, which controls the positioning of other components wrench **10** that comprise a ratcheting mechanism of the wrench **10** to support such functionality. Due to the ratcheting feature, the wrench **10**, may be particularly useful in confined areas that restrict movement by lessening or eliminating the need to reposition the wrench **10** on a fastener during a tightening or loosening operation.
(24) In this regard, FIG. **1** shows wrench **10**, according to some example embodiments, comprising a handle **15** and a head **20**. The head **20** may be disposed on a forward portion of the wrench **10**. The handle **15** may extend longitudinally away from the head **20** in a rearward direction to provide both a hand-grip and length from the rotational axis **205** of the wrench **10** to create wrench leverage for a user. According to some example embodiments, the handle **15** may be generally cylindrical in shape. The head **20** may have a top face **200** and a bottom face **210** for positional reference, where the top face **200** is opposite the bottom face **210** and the rotational axis **205** passes through the top face **200** and the bottom face **210**. The head **20** may broaden in width as the head **20** extends away from the handle **15**, and the head **20** may have a generally oval or ovoid shape. Within the head **20**, the wrench **10** may include a reversing ratcheting mechanism as further described below disposed within a head cavity within the head **20**. Various components of the wrench **10** may be formed or casted of metals, such as, for example steel or stainless steel.
(25) Additionally, the head **20** may include, extending from the bottom face **210**, a drive tang **110**. The drive tang **110** may be configured to couple to a work piece, such as socket, bit, or the like. In this regard, the drive tang **110** may be shaped to, for example, be received into a cavity of the work piece. According to some example embodiments, the drive tang **110** have a square cross-sectional shape. The drive tang **110** may be coupled to the ratcheting mechanism such that the drive tang **110** is configured to ratchet relative to the head **20** in a first rotational direction and apply a torque moving with the head **20** in a second rotational direction. As further, described below, the drive tang **110** may have an internal cavity that houses a locking member **115**, which may be a locking bearing. The locking member **115** may operate as a component of lockable detent **116** along with a locking member opening **112** in the drive tang **110**. The locking member **115** may be configured to engage with a work piece to lock the work piece to the drive tang **110** and the wrench **10**.
(26) In this regard, the wrench **10** may also include a release button **100** disposed on the top face **200**. The release button **100**, according to some example embodiments, may be movable or depressible into the top face **200** of the wrench **10**. The release button **100** may operate to control a

locking mechanism and, more specifically, the locking detent **116** for locking the work piece to the drive tang **110** via the locking member **115** as further described below. In short, when the release button **100** is not being acted upon, the release button **100** may default into a locked position, where the locking member **115** is maintained in an extended position, which would be engaged with a work piece if the work piece is installed on the drive tang **110**. However, the release button **100** may be depressed such that the release button **100** moves in a direction along or parallel the rotational axis **205** and into an unlocked position. When the release button **100** is disposed in the unlocked position, the locking member **115** may be permitted to move, for example, into the cavity in the drive tang **110**, thereby allowing a work piece to be removed from the drive tang **110**.

(27) According to some example embodiments, wrench **10** may include a slide lock member **120** that operates in cooperation with the release button **100** to prevent the unintentional release or separation of a work piece form the drive tang **110**. In this regard, according to some example embodiments, the slide lock member **120** may be disposed on the top face **200** of the wrench **10**. Similar to the release button **100**, the slide lock member **120** may be a moveable component. The slide lock member **120** may move, in a linear sliding fashion, on the top face **200** of the wrench **10** between an engaged position and a disengaged position. In the engaged position the slide lock member **120** may be in physical contact with the release button **100** in a manner that prevents the release button **100** from moving from the locked position to the unlocked position. Further, in the disengaged position, the slide lock member **120** may be disengaged from the release button **100**, and the release button **100** may be permitted to the move from the locked position to the unlocked position.

(28) Having described some of the external components of the wrench **10**, according to some example embodiments, FIG. **2** provides an exploded view of the various example components of the wrench **10**. In this regard, the wrench **10** may include a cover **33** that operates to maintain various components within a head cavity **37** of the head **20** and also protect internal components of the wrench **10** from dirt and debris. Accordingly, the cover **33** may be held in place by the locking ring **34**, which may be temporarily deformed to facilitate being installed in a channel in the cavity **37** of the head **20** to maintain the cover **33** on or adjacent to the bottom face **210** of the head **20**. The cover **33** may include an opening through which a drive tang **110** may pass and extend externally away from the head **20**. As mentioned above and further described below, the work piece may be locked onto the drive tang **110** via a lockable detect **116**. The lockable detent **116** may lock the work piece to the drive tang **110** through interaction with the lock member **115**. In this regard, the drive tang **110** may have an locking member opening **112** on one of the surfaces of the drive tang **110**. The locking member opening **112** may pass into a drive tang cavity **111**. The locking member **115** may be disposed in the drive tang cavity **111** may, in some instances, extend out of the locking member opening **112**. However, according to some example embodiments, locking member **115** may be too large to pass through the locking member opening **112**, but may still, in some instances, partially extend out of the locking member opening **112**.

(29) The drive tang **110** may be coupled to or integrated with a gear **30** comprising ratchet teeth encircling a body of the gear **30** on an external circular face of the gear **30**. The gear **30** may have a plurality of gear teeth **104** disposed about a circumference of the gear **30**, and further the drive tang **110** may be axially disposed on a base of the gear. In this regard, the gear **30** may have a circular shape that is elongated into a cylinder to increase the engagement surface area of the teeth for engagement with teeth of the pawl **29** which facilitate the ratcheting operation of the wrench **10**. To facilitate rotation of the gear **30** relative to the cover **33**, a washer **32** may be included therebetween. Additionally, to facilitate rotation of the gear **30** relative to the internal upper surface of the head cavity **37**, a washer **31** may be included therebetween.

(30) As mentioned above, the pawl **29** may have teeth that are configured to engage with the teeth of the gear **30** to facilitate the ratcheting functionality of the wrench **10**, thereby comprising components of the ratcheting mechanism **117** of the wrench **10**. The pawl **29** may be disposed in

the cavity **37** of the head **20**. The pawl **29** may be configured to move within the cavity **37** to permit a change in the ratcheting direction of the wrench **10**. In this regard, the reversing lever **25** may be configured to control the movement of the pawl **29** within the head **20**, and thereby control the ratcheting direction of the wrench **10**. The reversing lever **25**, may be disposed within the opening **38** in the head **20** and may engage with the pawl **29** via a slug **27** that applies a force on the pawl **29** via the spring **26** that is disposed between a body portion of the reversing lever **25** and the slug **27**. Additionally, to maintain the reversing lever **25** in a first position (e.g., for clockwise ratcheting) or a second position (e.g., for counter-clockwise ratcheting), a slug **23** may be engage with the body portion of the reversing lever **25**. Via the spring **24**, which is disposed between an internal surface of the cavity **37** and the slug **23**, a biasing force may be applied to the body portion of the reversing lever **25** that causes the reversing lever **25** to "jump" into one of the first position or the second position as the reversing lever **25** is pivoted. To facilitate rotation of the reversing lever **25** relative the top face **200** of the head **20**, a washer **28** maybe disposed therebetween.

(31) The wrench **10** may also include the release button **100** which may be coupled or integrated with a post **102**. The post **102** may extend through a top opening in the cavity **37** in the head **20** and form a portion of a lockable detent. The post **102** may extend from the bottom side of the release button **100** and may be disposed within a drive tang cavity **111** with the locking member **115**. As further described below, the post **102** may include features that operate to control the movement of the locking member **115** in response to movement of the release button **100** between the locked and unlocked positions. To maintain the release button **100** in a position that extends above the top face **200** of the head **20**, a release button spring **104** may urge the release button **100** into the extended, unlocked position. The release button spring **104** may be disposed within the drive tang cavity **111** and may be compressed between a ledge within the drive tang cavity **111** and a lower, spring lip of the release button **100**.

(32) According to some example embodiments, the slide lock member **120** and associated components may be disposed on a top face **200** of the head **20**. The slide lock member **210** may be secured to the top face **200** of the head **20** by securing pins **130**. The securing pins **130** may pass through securing pin openings in the slide lock member **120** and be secured (e.g., screwed) into pin holes **131** on the top face **200** of the head **20**. The securing pins **130** may have heads that may engage with a upper surface of the slide lock member **120** to hold the slide lock member **120** on the top face **200** of the head **20**. Additionally, a guide protrusion **132** may extend away from the top face **200** of the head **20** to engage a guide slot **127** (FIG. **3**B) to guide the sliding movement of the slide lock member **120**. Additionally, slide lock springs **140** may be may be disposed in spring slots **126** on the bottom side of the slide lock member **120** (FIG. **3**B) and may be positioned between the securing pins **130** and an end of the spring slots **126** to urge the slide lock member **120** into the engaged position.

(33) Having described the various individual components of the wrench **10**, FIGS. **3**A-**3**C will now be described which focus on the aspects of the slide lock member **120** and the securing pins **130**. In this regard, with reference to FIG. **3**A, a perspective top view of the slide lock member **120**, according to some example embodiments, is provided. According to some example embodiments, the slide lock member **120** is formed as a substantially planar component that includes a button opening **121**, securing pin openings **124**, and a thumb protrusion **123**.

(34) According to some example embodiments, the button opening **121** may be positioned such that the release button **100** may pass through the button opening **121**. As further described below, the edge of the button opening **121** may be configured to engage with the release button **100** underneath a locking lip **101** of the release button **100** to maintain the release button **100** in the locked position. However, via sliding movement of the slide lock member **120**, the edge of the button opening **121** may disengage from the locking lip **101** and permit the release button **100** to move through the button opening **121** into the unlocked position. According to some example embodiments, the edge of the button opening **121** may have a bevel **122** to facilitate sliding

engagement with the locking lip **101** (which may also have a bevel) of the release button **100**. As such, according to some example embodiments, the button opening **121** may be larger in area than an area than the release button **100** to permit the release button **100** to pass through the button opening **121**, when properly aligned. According to some example embodiments, the button opening **121** may be circular in shape. According to some example embodiments, the button opening **121** may be oval shaped or another elongated shape. Additionally, the button opening **121** may be, according to some example embodiments, closed such that the entire edge of the button opening **121** is disposed internal to the slide lock member **120** and does not extend to an external edge of the slide lock member **120**.

(35) The securing pin openings **124** may also be sized to facilitate the ability of the slide lock member **120** to slide relative to the head **20** and the release button **100**. In this regard, the securing pin openings **124** may be elongated along the sliding movement direction of the slide lock member **120** such that the securing pin openings **124** have first and second opposite ends. As such, the shape of the securing pin openings **124** may permit the slide lock member **120** to slide relative to the securing pins **130** which are affixed to the head **20** of the wrench **10**. According to some example embodiments, the edges of the securing pin openings **124** may include a bevel **125**.

(36) The thumb protrusion **123** may, according to some example embodiments, be disposed at a position forward of the button opening **121**. The thumb protrusion **123** may be a raised area that may be elongated to operate as a grip for a user's thumb. In this regard, as further described below, the thumb protrusion **123** may be the interface between the user and the slide lock member **120** to provide a means for sliding the slide lock member **120**, for example, into a disengaged position against the urging of the slide lock springs **140**. According to some example embodiments, the thumb protrusion **123** may be positioned adjacent to the button opening **121**, and thus the release button **100**, to permit the user to perform a two-step operation of sliding the slide lock member **120** and depressing the release button **100** with the same thumb.

(37) FIG. **3**B shows a bottom side of the slide lock member **120**. The bottom side of the slide lock member **120** may include features such as the spring slots **126** and the guide slot **127**. In this regard, the spring slots **126** may be elongated cavities that are configured to receive the slide lock springs **140**. According to some example embodiments, the spring slots **126** may be aligned with the securing pin openings **124** such that the securing pin openings **124** open into the spring slots **126**. In this regard, when assembled, a securing pin **130** may pass through a securing pin opening **124** and into the spring slot **126** to permit the slide lock spring **140** to engage directly with the securing pin **130** on one end of the slide lock spring **140** and with wall of the spring slot **126** at the other end of the slide lock spring **140**. In this regard, since the securing pin **130** is fixed in position relative to the head **20**, the slide lock spring **130** may use the securing pin **130** as leverage to urge the slide lock member **120** into the engaged position.

(38) Additionally, the guide slot **127** may also be an elongate cavity in the bottom surface of the slide lock member **120**. As mentioned above, the guide slot **127** may be sized and positioned to engage with the guide protrusion **132** disposed on the top face **200** of the head **20**. The engagement between the guide protrusion **132** and the guide slot **127** may operate to limit the movement of slide lock member **120** to a linear sliding movement along the length of the guide slot **127**, based on the shape of the guide slot **127**.

(39) Additionally, according to some example embodiments, the wrench **10** may also include casings **35**. The casings **35** may be configured to rest or on the top face **200** of the head **20** or be inserted into the pin holes **131** to rest on an ledge within the pin holes **131**, and the casing **35** may be configured to receive the securing pins **130**. In this regard, a securing pin **130** may pass through a channel of the casing **35** to engage with, for example, threading in the pin holes **131**. The casings **35** may operate, according to some example embodiments, as a stop or stand-off to control a depth of the securing pins **130** and prevent overtightening of the securing pins **130**, which may restrict or even prevent the sliding movement of the slide lock member **120**. The casing **35** may rest on the

top face **200** of the head **20** or a ledge within the pin hole **131**, and the head **133** of the securing pin **130** may be tightened against the casing **35** (rather than the slide lock member **120**) and the casing **35** may be sized accordingly.

(40) Further, according to some example embodiments, a casing **35** may include an upper head portion **39** that is wider than a lower body portion **40**. According to some example embodiments, the casings **35** may be installed through the slide lock member **120** such that the upper head portion **39** is seated above a bevel **125** or similarly positioned ledge within the securing pin opening **124** (as further described below with respect to FIGS. **3**A and **3**B). The upper head portion may therefore include a lower external ledge that may engage with the bevel **125** or similarly positioned ledge of the securing pin opening **124**. Because the lower body portion **40** of the casing **35** may rest on the backside of the head **20** or a ledge within the pin hole **131**, the securing pin **130** may be tightened into the channel of the casing **35** and onto an internal ledge within the channel of the casing **35**, with the head **133** being seated on the internal ledge in the upper head portion **39** of the channel in the casing **35**. With the securing pin **130** tightened, the slide lock member **120** may be held on the top face **200** of the head **20** by the upper head portion of the casing **35**, which in turn is held in place by the securing pin **130**. Because the casing **35** is prevented from being tightened onto the slide lock member **120**, the slide lock member **120** is free to slide relative to the casing **35** across the top face **200** of the head **20**, but cannot be moved away from the top face **200** with a component of movement in a direction perpendicular to the top face **200**. In addition to operating as a stop or stand-off for the securing pin **130**, the implementation of the casings **35** may also provide added rigidity and support to the securing pins **130** to limit or avoid bending or shifting of the securing pins **130** into improper engagement with the slide lock member **120** when the wrench **10** is subjected to an impact (e.g., when dropped from a high altitude).

(41) FIG. **3**C shows an example securing pin **130**. As shown, the securing pin **130** may include a head **133**, a shank **136**, and a thread **137**. The thread **137** may be sized to engage with corresponding threading in the pin holes **131** of the head **20**. The shank **136** may, according to some example embodiments, be a smooth portion of the securing pin **130** that may be configured to engage with the slide lock spring **140**, as described herein. The head **133** may be broader that the shank **136** to permit the head **133** to operate as a catch to secure the slide lock member **120** to the head **20** of the wrench **10**. In this regard, the head **133** may have a dimension that is larger than the width of the securing pin opening **124** to prevent the head **133**, for example, from passing through the securing pin opening **124**. In this regard, to permit the head **133** to sit flush on the top of the slide lock member **120** and still operate as a catch, the head **133** may have a bevel **135** that may correspond to the bevel **125** on the edge of the securing pin opening **124**. Additionally, the head **133** may include a drive **134** that facilitates engagement with a tool for tightening or loosening the securing pin **130** in the pin hole **131**.

(42) Having described physical and functional characteristics of various components of the wrench **10**, a description of the operation of the components will now be provided. In this regard, FIGS. **4**A-**4**D illustrate the configuration of various components of the wrench **10**, when the release button **100** is in the locked position and the slide lock member **120** is in the engaged position. With reference to FIGS. **4**A and **4**B, it can be seen that the slide lock member **120** has been slid forward (away from the handle **15**) to the forward-most position, which may be the engaged position. In this regard, as mentioned above, with the slide lock member **120** in the engaged position, the release button **100** is prevented from moving out of the locked position. It can be seen in FIG. **4**A, that the rear edge of the button opening **121** is engaged with the release button **100**, while a gap exists between the front edge of the button opening **121** and the release button **100**. As shown in FIG. **4**B, because the release button **100** is in the locked position, the locking member **115** is extended out of the opening the drive tang **110**.

(43) Now referencing FIG. **4**C, a zoomed view of the head **20** is provided with the slide lock member **120** in the engaged position and the release button **100** in the unlocked position. In this

regard, the dashed circle **204** indicates the position of the edge of the button opening **121**, which is obscured by the locking lip **101** of the release button **100**. As shown, due to the locking lip **101** extending away from a center of the release button **100** to create a void, the edge of the button opening **121** can slide under locking lip **101** into the void, thereby blocked the release button **100** form moving into the unlocked position.

(44) Additionally, as shown in FIG. **4**C, the securing pin openings **124** are also slid forward due to the slide lock member **120** being in the engaged position. As such, the securing pins **130** are located against the rear edge of the securing pin openings **124**. Further, the slide lock springs **140** can be seen though the securing pin openings **124** in positions engaged with the securing pins **130** and urging the slide lock member **120** forward into the engaged position.

(45) Now referencing FIG. **4**D, a cross-section view taken at A-A of FIG. **4**C of select components of the wrench **10** is shown, with the slide lock member **120** in the engaged position. In this regard, FIG. **4**D illustrates the positioning of the slide lock member **120**, the release button **100**, the post **102**, and the locking member **115** relative to the gear **30** and the drive tang **110**, when the slide lock member **120** is in the engaged position. The release button **100**, the post **102**, and the locking member **115** may be components of the lockable detent **116**.

(46) The slide lock member **120** is shown with a top surface being slid under the locking lip **101** of the release button **100** (which may have a bevel **107**). As such, in this position, the engagement between the locking lip **101** and the edge of the button opening **121** of the slide lock member **120** prevents the release button **100** from being depressed and moving from the locked position to the unlocked position. Additionally, the force provided by the release button spring **104** urges the release button **100** into the locked position due to the engagement of the release button spring **104** between the spring lip **105**, from which the post **102** extends, and a ledge within the drive tang cavity **111**.

(47) With the release button **100** in the locked position, the post **102** is positioned such that a shallow edge **104** of the post **102** is aligned with the locking member opening **112**. As such, the locking member **115**, which may be in the form of a spherical bearing, may be forced to extend out of the locking member opening **112** due to the engagement or positioning of the shallow edge **104** of the post **102**. As mentioned above, the locking member opening **112** may be smaller than the locking member **115**, and therefore, the locking member **115** may extend out of, but may not pass completely through the locking member opening **112**.

(48) Now with reference to the FIGS. **5**A to **5**E, the transition to the slide lock member **120** being in the disengaged position and the release button **100** being in the unlocked position will be described with respect to the various components of the wrench **10**. In this regard, FIG. **5**A provides a top view and FIG. **5**B provides a side view of the wrench **10** shown after the slide lock member **120** has be slid rearward (towards the handle) along the direction **201** (which may be a linear direction) into the disengaged position. The sliding movement of the slide lock member **120** may be constrained to a linear sliding movement by, for example, the shape of the securing pin openings **124** and the guide slot **127**. As can be seen in FIG. **5**A, the securing pin openings **124** have now shifted into a position where the securing pins **130** are engaged with a forward end of the securing pin openings **124**. According to some example embodiments, the engagement between the forward ends of the securing pin openings **124** and the securing pins **130** may operate as a stop to prevent further movement of the slide lock member **120** in the rearward direction. With the slide lock member **120** in the disengaged position, the button opening **121** is aligned with the release button **100** such that the edge of the button opening **121** is not disposed in engagement with the locking lip **101** of the release button **100**.

(49) Now with reference to FIG. **5**C, the slide lock member **120** has been slid in the direction **201**, and now the release button **100** has been depressed and moved in the direction **202**, which may be a linear direction. The sliding movement of the slide lock member **120** in the direction **201**, according to some example embodiments, may be perpendicular to the movement of the release

button **100** in the direction **202**. Additionally, the sliding movement of the slide lock member **120** may be a linear motion and the movement of the release button **100**, due to being pressed into the top face **200** of the head **20**, may also be a linear movement. Because the release button **100** has been depressed and moved into the unlocked position, the locking detent **116** comprising the locking member **115** may permit the locking member **115** to move out of the extended position to permit a work piece to be removed from the drive tang **110**.

(50) FIG. **5**D provides a zoomed, top view of the head **20** with the slide lock member **120** in the disengaged position and the release button **100** in the unlocked position. In this regard, it can be seen, as mentioned above, that the securing pin openings **124** have shifted rearward and the securing pins **130** are now disposed in an forward position within the securing pin openings **124** and engaged with the forward edges of the securing pin openings **124**. As such, the engagement between the securing pins **130** and the forward edges of the securing pin openings **124** may operate as a stop to prevent further movement of the slide lock member **120** in the rearward direction. Additionally, as shown in FIG. **5**D, the release button **100** is aligned with the button opening **121**. As such, the edge of the button opening **121** is no longer engaged with the locking lip **101** of the release button **100**. Therefore, the release button **100** is moveable (i.e., may be depressed) unencumbered by the slide lock member **120**.

(51) Now referencing FIG. **5**E, a cross-section view taken at B-B of FIG. **5**D of select components of the wrench **10** is shown, with the slide lock member **120** in the disengaged position and the release button **100** in the unlocked position. In this regard, FIG. **5**E illustrates the positioning of the slide lock member **120**, the release button **100**, the post **102**, and the locking member **115** relative to the gear **30** and the drive tang **110**, when the slide lock member **120** is in the disengaged position and the release button **100** is in the unlocked position.

(52) The slide lock member **120** is shown with a top surface of the slide lock member **120** being slid rearward such that the locking lip **101** of the release button **100** is no longer in engagement with the locking lip **101** and does not block the release button **100** from being depressed and moved into the unlocked position. As such, in this position, the button opening **121** and the release button **100** are aligned, and, at least a portion of the release button **100** may pass through the button opening **121** into the drive tang cavity **111** when depressed. Further, the release button spring **104** may be compressed between the spring lip **105** and the ledge within the drive tang cavity **111**.

(53) With the release button **100** in the unlocked position, the post **102** is positioned such that a deep edge **103** of the post **102** is aligned with the locking member opening **112**. As such, the locking member **115**, which may be in the form of a spherical bearing, may be permitted to move from the extended position into a retracted position, where, for example, the locking member **115** is completely or substantially disposed within the drive tang cavity **111** and no longer extends through the locking member opening **112**. As such, any work piece that was being locked in place due to engagement with the locking member **115**, is no longer in engagement with the locking member **115** and may be removed or separated from the drive tang **110**.

(54) FIGS. **6**A to **6**C provide illustrations of a user operating the wrench **10**, according to some example embodiments. In this regard, with reference to FIG. **6**A, a user's thumb **203** is attempting to push or depress the release button **100** with the slide lock member **120** in the engaged position, in an attempt to unlock the socket **190** from the drive tang **110**. Because the slide lock member **120** is engaged with the release button **100**, the release button **100** cannot be moved into the unlocked position, and therefore the socket **190** remains locked onto the wrench **10**.

(55) Now with reference to FIG. **6**B, the user's thumb **203** has engaged the thumb protrusion **123** and the user's thumb has pulled the slide lock member **120** rearward, towards the handle **15** in the direction **201**. As such, due to, for example, the stops formed by the securing pin openings **124** and the securing pins **130**, the slide lock member **120** may be prevented from sliding further rearward, and the release button **100** may be in alignment with the button opening **121**. Since the release button **100** has not yet been depressed, the lockable detent **116** is still operating to lock the socket

190 to the drive tang **110**.

(56) Now referencing FIG. **6**C, while maintaining the slide lock member **120** in the disengaged position (against the urging of the slide lock springs **140**), the user's thumb **203** now depresses release button **100**, thereby moving the release button **100** from the locked position to the unlocked position. As such, the lockable detent **116** permits the socket **190** to be separated from the drive tang **110** due to, for example, retraction of the locking member **115** into the drive tang cavity **111** and out of engagement with the socket **190**.

(57) As such, according to some example embodiments, a ratchet wrench is provided. The ratchet wrench may comprise a head comprising a top face and a bottom face, and a drive tang that extends from the bottom face of the head. In this regard, the drive tang may be shaped to engage a work piece. The ratchet wrench may further comprise a ratcheting mechanism disposed within a head cavity of the head and operably coupled to the drive tang. Additionally, the ratcheting mechanism may be configured to permit the drive tang to rotate relative to the head in a ratcheting rotational direction and prevent the drive tang from rotating relative to the head in a drive rotational direction. The ratchet wrench may further comprise a locking detent operably coupled to the drive tang, and a release button disposed on a top face of the head and configured to control operation of the locking detent. The release button may be movable along a first direction between a locked position, where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang. The ratchet wrench may further comprise a slide lock member configured to slide along a second direction between an engaged position and a disengaged position. In this regard, the second direction may be perpendicular to the first direction. Additionally, in the engaged position, the slide lock member may engage with the release button to prevent movement of the release button in the first direction from the locked position to the unlocked position, and, in the disengaged position, the slide lock member may be disengaged from the release button to permit movement of the release button in the first direction to move from the locked position to the unlocked position.

(58) Additionally, the ratchet wrench may further comprise a release button spring configured to urge the release button into the locked position, and a slide lock spring configured to urge the slide lock member into the engaged position. Additionally or alternatively, according to some example embodiments, the release button may comprises a locking lip, and the slide lock member may be positioned under the locking lip of the release button when the slide lock member is in the engaged position. Additionally or alternatively, the locking lip may comprise a beveled edge. Additionally or alternatively, according to some example embodiments, the slide lock member may comprise a button opening that is aligned with the release button when the slide lock member is in the disengaged position such that a portion of the release button moves through the button opening of the slide lock member when the slide lock member is in the disengaged position and the release button is moved from the locked position to the unlocked position. Additionally or alternatively, the slide lock member may comprise a securing pin opening and a guide slot. In this regard, a securing pin may pass through the securing pin opening in the slide lock member and may couple to the top face of the head. Further, the securing pin opening may be sized to permit the slide lock member to slide relative to the securing pin. The slide lock member may comprise a guide protrusion extending from the top face of the head. The guide protrusion may be configured to extend into the guide slot of the slide lock member to guide movement of the slide lock member along the second direction. Additionally or alternatively, the securing pin opening may be an elongate opening having a first end and an second end. Additionally, the engagement between the securing pin and the first end operates as a stop for slide lock member in the disengaged position. Additionally or alternatively, according to some example embodiments, the slide lock member may comprise a spring slot that is aligned with the securing pin, and the ratchet wrench may further comprise a slide lock spring disposed within the spring slot such that an end of the slide lock spring engages with the securing pin. In this regard, the slide lock spring may be configured to urge the slide lock

member into the engaged position by applying a force on the securing pin. Additionally or alternatively, according to some example embodiments, the slide lock member may comprise a thumb protrusion configured to be engageable with a thumb of a user to facilitate sliding of slide lock member from the engaged position to the disengage position. Additionally or alternatively, according to some example embodiments, the locking detent may comprise a post operably coupled to the release button such that the post moves with the release button along the first direction, and the post may be disposed within a drive tang cavity in the drive tang. The locking detent may also comprise a locking member disposed within the drive tang cavity and operably coupled to the post such that the movement of the post moves the locking member between an extended position, where a portion of the locking member extends out of an opening in the drive tang to lock the work piece to the drive tang, and a retracted position that permits the work piece to be removed from the drive tang.

(59) According to some example embodiments, a wrench is provided. The wrench may comprise a head comprising a top face and a bottom face, and a drive tang that extends from the bottom face of the head. In this regard, the drive tang may be shaped to engage a work piece. The wrench may further comprise a locking detent operably coupled to the drive tang and a release button disposed on a top face of the head and configured to control operation of the locking detent. The release button may be movable along a first linear direction between a locked position where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang. The wrench may further comprise a slide lock member configured to slide along a second linear direction between an engaged position and a disengaged position. In this regard, in the engaged position, the slide lock member may engage with the release button to prevent movement of the release button in the first linear direction from the locked position to the unlocked position, and, in the disengaged position, the slide lock member may be disengaged from the release button to permit movement of the release button in the first linear direction to move from the locked position to the unlocked position.

(60) Additionally or alternatively, according to some example embodiments, the wrench may comprise a release button spring configured to urge the release button into the locked position, and a slide lock spring configured to urge the slide lock member into the engaged position. Additionally or alternatively, according to some example embodiments, the release button comprises a locking lip, and the slide lock member is positioned under the locking lip of the release button when the slide lock member is in the engaged position. Additionally or alternatively, according to some example embodiments, the slide lock member may comprise a button opening that is aligned with the release button when the slide lock member is in the disengaged position such that a portion of the release button moves through the button opening of the slide lock member when the slide lock member is in the disengaged position and the release button is moved from the locked position to the unlocked position. Additionally or alternatively, according to some example embodiments, the slide lock member may comprise a securing pin opening and a guide slot. The wrench may also comprise a securing pin that passes through the securing pin opening in the slide lock member and couples to the top face of the head. The securing pin opening may be sized to permit the slide lock member to slide relative to the securing pin. Further, the wrench may comprise a guide protrusion extending from the top face of the head, and the guide protrusion may be configured to extend into the guide slot of the slide lock member to guide movement of the slide lock member along the second linear direction. Additionally or alternatively, according to some example embodiments, the securing pin opening may be an elongate opening having a first end and an second end, and engagement between the securing pin and the first end operates as a stop for slide lock member in the disengaged position. Additionally or alternatively, the slide lock member may be a spring slot that is aligned with the securing pin. In this regard, the wrench further comprises a slide lock spring disposed within the spring slot such that an end of the slide lock spring engages with the securing pin, and the slide lock spring is configured to urge the slide lock member into the engaged position

by applying a force on the securing pin.
(61) Many modifications and other embodiments of the inventions set forth herein will come to mind to one skilled in the art to which these inventions pertain having the benefit of the teachings presented in the foregoing descriptions and the associated drawings. Therefore, it is to be understood that the inventions are not to be limited to the specific embodiments disclosed and that modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although the foregoing descriptions and the associated drawings describe exemplary embodiments in the context of certain exemplary combinations of elements and/or functions, it should be appreciated that different combinations of elements and/or functions may be provided by alternative embodiments without departing from the scope of the appended claims. In this regard, for example, different combinations of elements and/or functions than those explicitly described above are also contemplated as may be set forth in some of the appended claims. In cases where advantages, benefits or solutions to problems are described herein, it should be appreciated that such advantages, benefits and/or solutions may be applicable to some example embodiments, but not necessarily all example embodiments. Thus, any advantages, benefits or solutions described herein should not be thought of as being critical, required or essential to all embodiments or to that which is claimed herein. Although specific terms are employed herein, they are used in a generic and descriptive sense only and not for purposes of limitation.

Claims

1. A ratchet wrench comprising: a head comprising a top face and a bottom face; a drive tang that extends from the bottom face of the head, the drive tang being shaped to engage a work piece; a ratcheting mechanism disposed within a head cavity of the head and operably coupled to the drive tang, the ratcheting mechanism being configured to permit the drive tang to rotate relative to the head in a ratcheting rotational direction and prevent the drive tang from rotating relative to the head in a drive rotational direction; a locking detent operably coupled to the drive tang; a release button disposed on a top face of the head and configured to control operation of the locking detent, the release button being movable along a first direction between a locked position where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang; and a slide lock member configured to slide along a second direction between an engaged position and a disengaged position, the second direction being perpendicular to the first direction; wherein, in the engaged position, the slide lock member engages with the release button to prevent movement of the release button in the first direction from the locked position to the unlocked position; and wherein, in the disengaged position, the slide lock member is disengaged from the release button to permit movement of the release button in the first direction to move from the locked position to the unlocked position; wherein the slide lock member comprises a securing pin opening and a guide slot; and wherein the ratchet wrench further comprises: a securing pin that passes through the securing pin opening in the slide lock member and couples to the top face of the head, the securing pin opening being sized to permit the slide lock member to slide relative to the securing pin; a guide protrusion extending from the top face of the head, the guide protrusion being configured to extend into the guide slot of the slide lock member to guide movement of the slide lock member along the second direction.
2. The ratchet wrench of claim 1 further comprising: a release button spring configured to urge the release button into the locked position; and a slide lock spring configured to urge the slide lock member into the engaged position.
3. The ratchet wrench of claim 1 wherein the release button comprises a locking lip; and wherein the slide lock member is positioned under the locking lip of the release button when the slide lock member is in the engaged position.
4. The ratchet wrench of claim 3, wherein the locking lip comprises a beveled edge.

5. The ratchet wrench of claim 3, wherein the slide lock member comprises a button opening that is aligned with the release button when the slide lock member is in the disengaged position such that a portion of the release button moves through the button opening of the slide lock member when the slide lock member is in the disengaged position and the release button is moved from the locked position to the unlocked position.
6. The ratchet wrench of claim 1 wherein the securing pin opening is an elongate opening having a first end and an second end; and wherein engagement between the securing pin and the first end operates as a stop for slide lock member in the disengaged position.
7. The ratchet wrench of claim 1, wherein the slide lock member comprises a spring slot that is aligned with the securing pin; and wherein the ratchet wrench further comprises a slide lock spring disposed within the spring slot such that an end of the slide lock spring engages with the securing pin; wherein the slide lock spring is configured to urge the slide lock member into the engaged position by applying a force on the securing pin.
8. The ratchet wrench of claim 1, wherein the slide lock member comprises a thumb protrusion configured to be engageable with a thumb of a user to facilitate sliding of slide lock member from the engaged position to the disengage position.
9. The ratchet wrench of claim 1, wherein the locking detent comprises: a post operably coupled to the release button such that the post moves with the release button along the first direction, the post being disposed within a drive tang cavity in the drive tang; a locking member disposed within the drive tang cavity and operably coupled to the post such that the movement of the post moves the locking member between an extended position, where a portion of the locking member extends out of an opening in the drive tang to lock the work piece to the drive tang, and a retracted position that permits the work piece to be removed from the drive tang.
10. A wrench comprising: a head comprising a top face and a bottom face; a drive tang that extends from the bottom face of the head, the drive tang being shaped to engage a work piece; a locking detent operably coupled to the drive tang; a release button disposed on a top face of the head and configured to control operation of the locking detent, the release button being movable along a first linear direction between a locked position where the locking detent locks the work piece to the drive tang and an unlocked position where the detent permits removal of the work piece from the drive tang; and a slide lock member configured to slide along a second linear direction between an engaged position and a disengaged position, the slide lock member comprising a button opening positioned such that the release button is able to pass through the button opening; wherein, in the engaged position, the slide lock member engages with the release button to prevent movement of the release button in the first linear direction from the locked position to the unlocked position; and wherein, in the disengaged position, the slide lock member is disengaged from the release button to permit movement of the release button in the first linear direction to move from the locked position to the unlocked position; wherein the slide lock member comprises a thumb protrusion that is an elongated raised area that operates as a grip for a user's thumb; wherein the thumb protrusion is positioned adjacent to the button opening and the release button to permit the user to perform a two-step operation of sliding the slide lock member and depressing the release button with a same thumb of the user.
11. The wrench of claim 10 further comprising: a release button spring configured to urge the release button into the locked position; and a slide lock spring configured to urge the slide lock member into the engaged position.
12. The wrench of claim 10 wherein the release button comprises a locking lip; and wherein the slide lock member is positioned under the locking lip of the release button when the slide lock member is in the engaged position.
13. The wrench of claim 12, wherein the button opening that is aligned with the release button when the slide lock member is in the disengaged position such that a portion of the release button moves through the button opening of the slide lock member when the slide lock member is in the

disengaged position and the release button is moved from the locked position to the unlocked position.
14. The wrench of claim 10, wherein the slide lock member comprises a securing pin opening and a guide slot; and wherein the wrench further comprises: a securing pin that passes through the securing pin opening in the slide lock member and couples to the top face of the head, the securing pin opening being sized to permit the slide lock member to slide relative to the securing pin; a guide protrusion extending from the top face of the head, the guide protrusion being configured to extend into the guide slot of the slide lock member to guide movement of the slide lock member along the second linear direction.
15. The wrench of claim 14 wherein the securing pin opening is an elongate opening having a first end and a second end; and wherein engagement between the securing pin and the first end operates as a stop for slide lock member in the disengaged position.
16. The wrench of claim 14, wherein the slide lock member comprises a spring slot that is aligned with the securing pin; and wherein the wrench further comprises a slide lock spring disposed within the spring slot such that an end of the slide lock spring engages with the securing pin; wherein the slide lock spring is configured to urge the slide lock member into the engaged position by applying a force on the securing pin.